

Professional Experience

- **Program Associate, International Forum for Environment, Sustainability & Technology, Delhi** [July'25-Present]
 - Compiled and ran **WRF** from scratch on **cloud HPC** for two Indian cities, reaching **RMSE 3.25** for 2 m air temperature
 - Developed **Digital Urban Climate Twins** chaining **WRF, PALM-4U, CEA** to test urban heat island interventions
 - Automated **WUDAPT** training area generation from urban morphometry, yielding **LCZ** maps at **75–80% accuracy**
 - Benchmarked **XGBoost** and **CNN** on identical inputs to attribute per-class LCZ performance
 - Built pipelines for **PALM-4U static driver generation** and urban data assembly for **geo-analytical workflows**
- **Research Assistant, Royal Danish Academy, Copenhagen** [Aug'21-Dec'22]
 - Developed 4 cost-effective **IoT PM sensors** outperforming commercial models, deployed and monitored remotely
 - Validated CFD output against measured **PET** and **UTCI** for experimental solar homes, for **disease vector control**
 - Built mesh based **RANS** CFD models (**OpenFOAM, ANSYS Fluent**) simulating wind flow for architectural design
- **Project Manager, Fairfest Media Limited, Mumbai** [June'18-July'19]
 - Spearheaded a multi-city team, delivering strategic insights for partners in Urban sector producing a news digest
 - Conducted **market segmentation, business, and financial modeling** to position the trade show commercially

Projects

- **Policy Intelligence Decision Support System for Spatial Governance** [Oct'25–Present]
 - Architecture for **policy intelligence decision engine** converting **policy queries** into spatial analytical tasks
 - **Semantic policy model** linking policy goals, regulatory constraints, spatial indicators, and computational workflows
 - Built an **explainable geo-analysis** by integrating heterogeneous data for **ontology-driven policy knowledge graph**
 - Implementing a **hybrid reasoning framework** to support scenario evaluation and policy trade-off analysis
- **PSInSAR for Urban Land subsidence monitoring** [Jul'24-June'25]
 - Conducting a **time series analysis** of urban land subsidence using Sentinel-1 and PSInSAR techniques in GIS
 - Optimising the **cloud computing** process of large-scale Microwave data on the Alaska Satellite Facility's platform
 - Developing advanced **Open Source** geoinformatics methods and tools for **urban** and **infrastructure resilience**
- **Flood Mapping using SAR and Optical Data** [March'24-April'24]
 - Enhanced flood segmentation model performance by optimising hyper-parameters improving to **82% accuracy**
 - Implemented **Attention U-Net** on optical data to determine urban and forest area for precise flood detection
 - Integrated with SAR in **Google Earth Engine** enabling prediction from Sentinel-1 images at **F1-score of 0.823**
- **Ontology and Sensor Location of Water Dist. Network** [Nov'23-March'24]
 - Optimized smart water meter placement using QGIS, K-means clustering and EPANET predicting water demand
 - Developed an ontology for water quality sensors with OWL and Protégé, enabling **semantic search** capabilities
 - Created an **IoT system** with water quality sensors, enabling real-time monitoring of the water distribution system
- **Acoustic Modeling game engine-based urban Digital Twin** [June'20-July'20]
 - Developed sound propagation models integrating **raytracing techniques** and environmental acoustics principles
 - Optimized design of **cellulose-based bio-material acoustic panels** for noise reduction in urban design
- **Urban Analytic: Predictive Modeling on Synthetic Data** [Sep'24-Nov'24]
 - Developed predictive models using **synthetic geospatial urban data** that comply with theoretical metrics
 - Applied model explainability methods (**SHAP values**) to interpret feature importance, for urban transportation

Education

- **M.Tech Geoinformatics, CSRE, Indian Institute of Technology Bombay, Mumbai** [Aug'23-Aug'25]
 - Specialized in spatial analytics, enhancing data-driven decision-making for urban and environmental applications
 - Integrated IoT and digital twin frameworks to develop interactive geospatial solutions to support urban projects
- **MA Computation in Architecture, CITA, Royal Danish Academy, Copenhagen** [Aug'19-July'21]
 - Developed **Computational Fluid Dynamics (CFD) models** for architectural space, to enhance design comfort
 - Applied **meshless particle-based CFD modeling** for urban flood resilience, simulating flood mitigation design
 - Researched cellulose 3D printing for **Fabrication Information Modeling** using image processing and correlation
- **Bachelor of Architecture, KRVIA, University of Mumbai, Mumbai** [Jun'12-Jun'17]
 - Designed A-Temporal Wetland project for Loktak, integrating technology appropriation with floating hut design
 - Design was shortlisted for the City and Water awards, the work led to a research fellowship opportunity

Technical Skills

Urban Climate & Atmospheric Modelling: WRF | PALM-4U (LES) | CEA | ENVI-met | WUDAPT / LCZ
Simulation & Digital Twins: OpenFOAM | ANSYS Fluent | RANS / CFD | Unity
GIS & Remote Sensing: QGIS | ArcGIS | PostGIS | Google Earth Engine | SNAP / PSInSAR
Programming & Computing: Python | R | SQL | C/C++ | Linux | Cloud HPC
AI & GeoAI: XGBoost | CNN | Attention U-Net | Explainable AI (SHAP) | GeoAI
Semantic & Knowledge Modelling: OWL | RDF | Protégé | Ontology Design | Neo4j
Instrumentation & Languages: Low-cost IoT sensing, field deployment | English | Hindi | Marathi